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GEMINI: **G**raphene-Foil **E**lectron-Emission **M**easurement **I**nvestigation for **N**ovel **I**nstrumentation

In space physics, amorphous carbon foils are a standard tool in particle-detecting space instruments because they release secondary electrons when struck by an incident particle. These secondary electrons can be used to make coincidence or time-of-flight (TOF) measurements, which are used by scientists to deduce properties of the incident particle such as mass, species, and energy. However, little is known, beyond a handful of published experiments, about exactly how the process of secondary electron emission works and how foil thickness and composition affect the fluence and energy of secondary electrons.

To address this gap, this proposal aims to develop an ultra-high-vacuum (UHV) compatible scientific instrument that examines the effect of foil thickness and composition on secondary electron emission in a laboratory setting. This laboratory instrument will position a carbon-based foil in a particle beam to measure incident and secondary particles and separate the transmitted particles by charge state. The instrument will be designed based on an existing device built by students in Princeton's 2023-2024 AST250/251 Space Physics Lab class, which was used to image carbon foils in the path of an ion beam and quantify the number of holes and wrinkles (defects) in the foil and their effect on its performance . The new instrument created as part of this research, GEMINI – which stands for Graphene-Foil Electron-Emission Measurement Investigation for Novel Instrumentation – will hold at least three carbon-based foils, concentrically partition charges, and measure secondary electron emission from the foils, allowing the instrument operator to characterize and perform side-by-side comparisons of amorphous carbon and few-layer-graphene (FLG) foils under the same particle beam. The end scientific goal of this research is to create an empirical model that infers foil thickness and composition based on the fluence and energy of secondary electrons emitted from the foil when it is placed in an incident particle beam.

In current literature it is hypothesized that the thinner the foil, the more sensitive a space instrument will be to detecting higher mass and lower energy particles, so a graphene foil, as a single layer of carbon, is expected to perform significantly better than amorphous carbon. However, existing experimentally-based findings in the literature do not support that theory and show that graphene only has marginal improvements over amorphous carbon. In addition to the carbon layer thickness, it is hypothesized that there is a "crud layer" of water vapor and other trace elements that accumulates on both types of carbon-based foils which significantly increases the thickness and negates the supposed benefits of a thinner foil. The development of GEMINI, will therefore, hold tremendous scientific value. It will mark the first ever study of secondary electron emission as a function of graphene foil thickness and composition and will allow future researchers to reliably characterize different carbon-based foils with varying thicknesses and surface compositions in a laboratory setting, in search for the optimal candidates for space flight.

As previously mentioned, GEMINI will build upon an apparatus first designed by the 2023-2024 AST250/251 class. The class's instrument uses three linear actuators to move a pinhole and carbon foil in front of a microchannel plate (MCP) imager so that when the device is placed in line with a beam of particles in the Space Physics Lab's vacuum chamber, the MCP creates a 2D spatial map of particle counts. It also has a metal gate (consisting of 2 parallel rods) that can be energized to high voltage and serves as an electrostatic analyzer (ESA), deflecting particles based on their energy and/or charge. This creates a visual distribution on the MCP which can be used to quantify how many of the particles are negatively charged, positively charged, or neutral, which in turn provides information on the composition of the foil.

Building upon the AST250/251 design, to meet the goals of this research, a channel electron multiplier (CEM) and new electro-optical components will be added to the design to measure the energy and fluence of secondary electrons emitted from the foil. CEMs multiply 1 electron to ~107-108 electrons, essentially creating a cascade of electrons with enough charge to be turned into a detectable signal. Adding a CEM directly contributes to the science goals of GEMINI, as it allows us to investigate how the foil thickness and composition affect the energy and fluence of secondary electron emission as the incident particles interact with the foil. GEMINI also fixes some mechanical flaws with the AST250/251 design, primarily related to the linear actuation method, which cannot precisely control the position of the pinhole and carbon foils as intended and is prone to getting stuck, making the device unreliable. The entire device is also tilted at a 45-degree angle to facilitate retraction of the linear actuators which are only controllable in the extending direction. However, this poses challenges for experimental reproducibility and needs to be re-examined.

For this project, SolidWorks CAD software will be used to mechanically design GEMINI and files from SolidWorks will be exported to Ansys for analysis and SIMION for electro-optical simulations. From an educational development perspective, this independent research is an opportunity to improve CAD skills, specifically in Design for Manufacturing (DFM). This project is also an opportunity to learn how to perform simulations in Ansys, including, but not limited to structural FEA and thermal simulations. Lastly, the project will also provide experience with SIMION, specifically in verifying simulation results by comparing them to collected data.

In total, the project duration spans both the Fall 2025 and Spring 2026 semesters. For Fall 2025 the requirements for this project are to (1) complete the redesign of the existing AST250/251 class instrument, (2) achieve a 15% reduction in mass without increasing the volume of the instrument, and (3) test the instrument and achieve a reliability of 10 cumulative hours in a beamline without a fault. In Spring 2026, with GEMINI built, the focus of the research will shift to taking scientific measurements of amorphous carbon foils and FLGs and creating the model that connects the foil's thickness and composition properties to the data collected by the instrument. The foils will then be taken to Princeton's Imaging and Analysis Center (IAC) for foil thickness and composition measurements, in order to quantify the agreement between the model and IAC measurements, with a goal of achieving agreement to within 10%. In terms of project management, this project involves a weekly advisor-student meeting to ensure continuous satisfactory progress, attendance at the Space Physics Lab weekly standup at 8AM on the first day of the week, and scheduled time to work in the lab from 1PM-5PM on Thursdays and 8:30AM-5PM on Fridays.